Praveen Madupu
Mb: +91 98661 30093
Sr. Database Administrator

Real-time scenario–based interview questions for **MariaDB Administration, Performance Tuning & Optimization:**

🟢 MariaDB Administration, Performance Tuning & Optimization – 500 Scenario-Based Questions

🔷 Basic Level (0–2 years) – 1–100

Installation & Configuration

1. How do you install MariaDB on Linux and Windows for production?
2. How do you secure a new MariaDB installation?
3. How do you configure MariaDB to listen on specific IP addresses?
4. How do you change the default data directory safely?
5. How do you enable SSL/TLS for client connections?
6. How do you configure MariaDB to use a custom port?
7. How do you enable query logging for slow queries?
8. How do you configure the MariaDB configuration file (my.cnf) for production?
9. How do you enable the InnoDB storage engine and tune its cache size?
10. How do you configure MariaDB to start automatically on system boot?

User Management & Security

11. How do you create users with role-based privileges?
12. How do you grant read-only access to reporting users?
13. How do you revoke privileges safely in production?
14. How do you enforce strong password policies for MariaDB users?
15. How do you detect unauthorized login attempts?
16. How do you audit user activities in MariaDB?
17. How do you integrate LDAP authentication with MariaDB?
18. How do you rotate passwords for MariaDB users without downtime?
19. How do you enforce network restrictions using bind-address?
20. How do you configure MariaDB for multi-factor authentication?

Database & Table Management

21. How do you create a new database and verify it exists?
22. How do you safely drop a production database?
23. How do you create a table with proper indexing for performance?
24. How do you alter a table to add a new column without locking it?
25. How do you rename a table in a production environment?
26. How do you set table constraints for data integrity?
27. How do you analyze table size and storage usage?
28. How do you monitor table growth over time?
29. How do you implement partitioning for large tables?
30. How do you convert a table from MyISAM to InnoDB safely?

Backup & Recovery

31. How do you take a logical backup using mysqldump?
32. How do you restore a database from a backup?
33. How do you implement incremental backups using mariabackup?
34. How do you schedule automated backups without impacting performance?
35. How do you backup only specific tables?
36. How do you verify the integrity of a backup?
37. How do you compress backups to save storage space?
38. How do you restore a single table from a full backup?
39. How do you perform point-in-time recovery using binary logs?
40. How do you implement backup strategies for replication setups?

Query Performance – Basics

41. How do you detect slow queries using the slow query log?
42. How do you use EXPLAIN to analyze query execution plans?
43. How do you optimize queries performing full table scans?
44. How do you detect queries causing high CPU usage?
45. How do you create indexes to speed up frequent queries?
46. How do you monitor query patterns over time?
47. How do you prevent queries from returning excessive rows?
48. How do you detect inefficient JOIN operations?
49. How do you optimize queries on columns with large text data?
50. How do you rewrite queries to avoid subquery performance issues?

Indexing Basics

51. How do you create single-column indexes?
52. How do you create multi-column (compound) indexes?
53. How do you detect unused indexes?
54. How do you remove redundant indexes safely?
55. How do you create unique indexes to prevent duplicates?
56. How do you implement fulltext indexes for text search?
57. How do you create partial or prefix indexes for large text fields?
58. How do you monitor index usage for optimization?
59. How do you rebuild fragmented indexes?
60. How do you balance indexing for read-heavy vs write-heavy workloads?

Memory & Storage

61. How do you monitor InnoDB buffer pool usage?
62. How do you tune innodb_buffer_pool_size for optimal performance?
63. How do you detect memory pressure in MariaDB?
64. How do you adjust innodb_log_file_size for write-heavy workloads?
65. How do you monitor disk space per database?
66. How do you detect table fragmentation and reclaim space?

Praveen Madupu
Mb: +91 98661 30093
Sr. Database Administrator

67. How do you use OPTIMIZE TABLE safely in production?
68. How do you configure multiple data directories for large storage?
69. How do you monitor disk I/O performance?
70. How do you tune InnoDB for high concurrency workloads?

Replication & High Availability

71. How do you set up master-slave replication?
72. How do you monitor replication lag?
73. How do you promote a slave to master safely?
74. How do you handle replication failures due to network issues?
75. How do you add a new slave to an existing replication setup?
76. How do you remove a failed slave safely?
77. How do you configure read-only slaves for reporting?
78. How do you detect replication conflicts or errors?
79. How do you implement semi-synchronous replication?
80. How do you verify replication consistency across nodes?

Monitoring & Logging

81. How do you enable and configure the general query log?
82. How do you enable and analyze the slow query log?
83. How do you monitor connections per user?
84. How do you detect queries causing table locks?
85. How do you monitor index usage and effectiveness?
86. How do you detect high CPU usage from queries?
87. How do you monitor disk I/O per table or database?
88. How do you integrate MariaDB logs with ELK/Prometheus?
89. How do you detect frequent deadlocks and resolve them?
90. How do you configure log rotation and retention policies?

Security Basics

91. How do you enforce SSL connections for all users?
92. How do you audit user login attempts?
93. How do you revoke unused privileges safely?
94. How do you implement encryption for data at rest?
95. How do you detect brute-force login attempts?
96. How do you implement firewall rules for MariaDB access?
97. How do you monitor for privilege escalation attempts?
98. How do you rotate server SSL certificates safely?
99. How do you secure the MariaDB configuration file against unauthorized changes?
100. How do you implement multi-factor authentication for database users?

Praveen Madupu
Mb: +91 98661 30093
Sr. Database Administrator

🟡 Intermediate Level (3–6 years) – 101–200

Query Performance & Optimization

101. How do you optimize queries with multiple JOINs?
102. How do you rewrite subqueries for performance improvement?
103. How do you analyze EXPLAIN output to detect bottlenecks?
104. How do you optimize queries on columns with NULL values?
105. How do you monitor and tune queries causing temporary table usage?
106. How do you optimize queries using covering indexes?
107. How do you detect queries performing filesort operations?
108. How do you reduce disk I/O during heavy SELECT operations?
109. How do you optimize GROUP BY queries for large datasets?
110. How do you handle large result sets efficiently?
111. How do you tune ORDER BY queries to avoid unnecessary sorting?
112. How do you optimize queries using LIMIT and OFFSET?
113. How do you use query hints for better execution plans?
114. How do you detect and resolve deadlocks in concurrent queries?
115. How do you optimize queries on VARCHAR and TEXT fields?
116. How do you implement partition pruning for performance?
117. How do you monitor long-running transactions?
118. How do you analyze query cache hit ratios?
119. How do you optimize queries with IN or OR conditions?
120. How do you design queries to minimize lock contention?

Indexing Intermediate

121. How do you monitor index usage over time?
122. How do you remove unused indexes without impacting performance?
123. How do you rebuild indexes for large tables without downtime?
124. How do you optimize indexes for read-heavy workloads?
125. How do you optimize indexes for write-heavy workloads?
126. How do you implement composite indexes for multi-column searches?
127. How do you detect and prevent duplicate indexes?
128. How do you use EXPLAIN to check index usage?
129. How do you optimize FULLTEXT searches using indexes?
130. How do you implement spatial indexes for GIS data?

Replication Intermediate

131. How do you monitor slave replication lag?
132. How do you detect broken replication chains?
133. How do you handle failover in a multi-slave environment?
134. How do you promote a slave safely in production?
135. How do you implement replication for high availability?
136. How do you optimize replication for low-latency networks?

Praveen Madupu
Mb: +91 98661 30093
Sr. Database Administrator

137. How do you prevent replication conflicts during schema changes?
138. How do you monitor semi-synchronous replication?
139. How do you troubleshoot replication errors using logs?
140. How do you detect delayed replication due to heavy writes?

Backup & Recovery Intermediate

141. How do you implement incremental backups with minimal downtime?
142. How do you monitor backup job performance?
143. How do you restore specific tables from full backups?
144. How do you perform point-in-time recovery using binary logs?
145. How do you automate backup verification?
146. How do you implement backup rotation policies?
147. How do you backup large databases efficiently?
148. How do you monitor backup storage usage?
149. How do you integrate backups with cloud storage?
150. How do you ensure consistent backups across replication nodes?

Memory & Storage Intermediate

151. How do you monitor InnoDB buffer pool usage for optimization?
152. How do you tune InnoDB log files for write-heavy workloads?
153. How do you detect and resolve memory pressure issues?
154. How do you optimize temporary table usage?
155. How do you analyze table fragmentation and reclaim space?
156. How do you monitor disk I/O performance in production?
157. How do you tune InnoDB for high-concurrency workloads?
158. How do you optimize queries to reduce disk usage?
159. How do you monitor and tune memory allocation for sorts?
160. How do you implement table partitioning for large tables?

Query Performance & Optimization (161–200)

161. How do you optimize queries using multiple indexes efficiently?
162. How do you analyze EXPLAIN output to detect temporary tables?
163. How do you tune queries performing full table scans?
164. How do you optimize JOIN queries with large tables?
165. How do you detect queries causing high lock contention?
166. How do you optimize GROUP BY queries on large datasets?
167. How do you tune ORDER BY queries to minimize disk I/O?
168. How do you optimize queries using LIMIT and OFFSET for pagination?
169. How do you rewrite subqueries as JOINs for better performance?
170. How do you use query hints to influence execution plans?
171. How do you detect queries causing high CPU usage?
172. How do you optimize queries with OR conditions?
173. How do you tune queries with IN clauses for large datasets?

174. How do you reduce disk usage during heavy SELECT operations?
175. How do you optimize queries on VARCHAR and TEXT columns?
176. How do you detect inefficient LIKE or REGEXP queries?
177. How do you implement query caching for frequently accessed data?
178. How do you monitor long-running transactions in production?
179. How do you optimize queries for partitioned tables?
180. How do you detect and resolve deadlocks in concurrent queries?

Indexing Advanced (181–220)

181. How do you implement composite indexes for multi-column queries?
182. How do you detect unused or redundant indexes?
183. How do you rebuild fragmented indexes without downtime?
184. How do you optimize indexes for read-heavy workloads?
185. How do you optimize indexes for write-heavy workloads?
186. How do you implement FULLTEXT indexes for efficient text search?
187. How do you implement spatial indexes for GIS data?
188. How do you monitor index usage and performance over time?
189. How do you detect index-related performance bottlenecks?
190. How do you tune index selection for complex queries?
191. How do you balance indexing overhead vs query performance?
192. How do you optimize indexes for JOIN queries?
193. How do you monitor and optimize InnoDB secondary indexes?
194. How do you detect index contention in high-concurrency environments?
195. How do you optimize indexes for foreign key lookups?
196. How do you optimize indexing for range queries?
197. How do you implement partial indexes for selective queries?
198. How do you monitor index rebuild impact on performance?
199. How do you design indexes for multi-tenant applications?
200. How do you implement indexing strategies for large tables?
201. How do you use prefix indexes for large VARCHAR columns?
202. How do you monitor slow queries caused by missing indexes?
203. How do you implement automatic index recommendations using MariaDB tools?
204. How do you detect index fragmentation affecting query speed?
205. How do you optimize indexes for queries with multiple JOINs?
206. How do you implement covered indexes to avoid table scans?
207. How do you detect and remove duplicate indexes?
208. How do you optimize queries using index hints?
209. How do you analyze and improve index selectivity?
210. How do you implement indexes for foreign key relationships?
211. How do you optimize FULLTEXT searches with proper indexing?
212. How do you monitor the impact of new indexes on write performance?
213. How do you optimize spatial queries using SPATIAL indexes?
214. How do you analyze and tune composite index order?

215. How do you prevent index bloat in large tables?
216. How do you monitor index usage in replication scenarios?
217. How do you implement indexes for partitioned tables?
218. How do you optimize multi-column queries using index coverage?
219. How do you detect and fix queries that ignore indexes?
220. How do you implement indexes to optimize reporting queries?

Replication & High Availability (221–260)

221. How do you configure master-slave replication in MariaDB?
222. How do you monitor replication lag for performance?
223. How do you promote a slave to master safely?
224. How do you detect broken replication chains?
225. How do you handle replication conflicts?
226. How do you implement semi-synchronous replication?
227. How do you optimize replication for high-write workloads?
228. How do you monitor replication error logs?
229. How do you add new slaves to an existing replication setup?
230. How do you remove a failed slave safely?
231. How do you implement replication for disaster recovery?
232. How do you configure read-only slaves for reporting?
233. How do you detect replication delays due to network latency?
234. How do you monitor binary log size and growth?
235. How do you perform backup and restore in a replicated environment?
236. How do you configure failover monitoring in replication?
237. How do you detect and handle skipped events in replication?
238. How do you optimize replication with multiple slaves?
239. How do you monitor replication consistency between nodes?
240. How do you implement automated failover in MariaDB replication?
241. How do you detect replication bottlenecks in high-load scenarios?
242. How do you synchronize new slaves efficiently?
243. How do you implement replication monitoring dashboards?
244. How do you handle replication with delayed slaves for PITR?
245. How do you detect replication conflicts during schema changes?
246. How do you configure replication filters for selective databases?
247. How do you monitor replication threads and performance?
248. How do you optimize replication performance across data centers?
249. How do you perform switchover from master to slave with minimal downtime?
250. How do you detect replication inconsistencies in large datasets?
251. How do you automate failover testing for replication setups?
252. How do you optimize replication for network-constrained environments?
253. How do you configure replication for multi-master setups (MariaDB Galera)?
254. How do you monitor Galera cluster health and performance?
255. How do you detect quorum loss in Galera clusters?

256. How do you handle node desynchronization in Galera clusters?
257. How do you optimize Galera replication for write-intensive workloads?
258. How do you perform rolling upgrades in a Galera cluster?
259. How do you monitor Galera cluster transaction certification performance?
260. How do you handle network partition in Galera clusters?

Backup & Recovery Advanced (261–300)

261. How do you implement point-in-time recovery using binary logs?
262. How do you schedule automated backups for minimal downtime?
263. How do you verify backup integrity automatically?
264. How do you restore a single table from a full backup?
265. How do you implement incremental backups with mariabackup?
266. How do you monitor backup performance in production?
267. How do you restore from incremental backups efficiently?
268. How do you integrate backups with cloud storage?
269. How do you compress backups to save storage space?
270. How do you implement backup retention policies?
271. How do you handle backup for replication slaves?
272. How do you test disaster recovery procedures regularly?
273. How do you implement encrypted backups?
274. How do you optimize backup processes for large databases?
275. How do you monitor backup logs for errors?
276. How do you restore MariaDB in a high-availability environment?
277. How do you perform backup while replication is running?
278. How do you automate backup verification checks?
279. How do you schedule off-peak backups without impacting production?
280. How do you implement multi-location backup strategies?
281. How do you optimize backup scripts for incremental backups?
282. How do you restore a corrupted database from backup?
283. How do you handle backups during schema changes?
284. How do you monitor disk usage for backups?
285. How do you implement hot backups with mariabackup?
286. How do you detect incomplete backups?
287. How do you implement backup encryption in transit and at rest?
288. How do you test point-in-time recovery regularly?
289. How do you restore partial data without affecting other tables?
290. How do you schedule automated offsite backups?
291. How do you integrate backups with monitoring dashboards?
292. How do you manage backup for multi-database servers?
293. How do you detect and fix corrupted backup files?
294. How do you handle backup in sharded or partitioned tables?
295. How do you restore Galera cluster nodes using backup?
296. How do you monitor backup execution time and performance?

297. How do you implement incremental backup verification?
298. How do you optimize disk I/O during backups?
299. How do you integrate backups with disaster recovery drills?
300. How do you implement version-controlled backup strategies?

Advanced Level – 301–500

Memory, Storage & Performance Tuning (301–340)

301. How do you monitor InnoDB buffer pool usage and hit ratio?
302. How do you tune innodb_buffer_pool_size for high-concurrency workloads?
303. How do you monitor disk I/O to detect bottlenecks?
304. How do you optimize innodb_log_file_size for write-intensive applications?
305. How do you detect memory pressure in MariaDB?
306. How do you optimize temporary table usage for large queries?
307. How do you monitor innodb_flush_log_at_trx_commit for performance vs durability?
308. How do you detect table fragmentation and reclaim space efficiently?
309. How do you monitor disk space usage per database and table?
310. How do you optimize queries to minimize disk usage and I/O?
311. How do you tune InnoDB redo log files for write-heavy workloads?
312. How do you monitor innodb_io_capacity and innodb_io_capacity_max?
313. How do you optimize queries to reduce memory footprint?
314. How do you detect slow queries consuming excessive buffer pool memory?
315. How do you optimize join queries to avoid temporary tables on disk?
316. How do you monitor and tune table cache usage (table_open_cache)?
317. How do you detect and resolve lock contention on large tables?
318. How do you optimize buffer pool usage for multiple databases?
319. How do you monitor page eviction events in InnoDB?
320. How do you optimize sorting operations to prevent disk spills?
321. How do you tune tmp_table_size and max_heap_table_size?
322. How do you detect queries causing swap usage?
323. How do you optimize multi-column queries for memory efficiency?
324. How do you monitor disk I/O during backup and restore operations?
325. How do you optimize large table scans with minimal buffer pool impact?
326. How do you tune InnoDB doublewrite buffer for high throughput?
327. How do you monitor and optimize InnoDB adaptive hash index?
328. How do you detect queries causing excessive redo log writes?
329. How do you optimize queries using covering indexes to reduce memory usage?
330. How do you monitor buffer pool resizing impact on performance?
331. How do you optimize aggregation queries for large datasets?
332. How do you tune innodb_flush_neighbors for SSD storage?
333. How do you detect and resolve contention on InnoDB insert buffers?
334. How do you monitor temporary disk usage for joins and sorts?
335. How do you optimize replication to reduce memory pressure on slaves?

Praveen Madupu
Mb: +91 98661 30093
Sr. Database Administrator

336. How do you detect and tune queries causing buffer pool thrashing?
337. How do you optimize large update/delete operations to reduce I/O spikes?
338. How do you tune InnoDB adaptive flushing for mixed workloads?
339. How do you monitor and optimize auto-increment contention in InnoDB?
340. How do you detect queries that cause excessive undo log generation?

Advanced Query Optimization (341–380)

341. How do you rewrite subqueries to improve performance?
342. How do you optimize multi-table joins with indexes?
343. How do you detect queries causing temporary table usage?
344. How do you optimize GROUP BY and ORDER BY queries?
345. How do you implement query caching for frequent reports?
346. How do you optimize DISTINCT queries for large tables?
347. How do you detect queries performing full table scans unnecessarily?
348. How do you tune queries using EXPLAIN FORMAT=JSON?
349. How do you optimize queries with multiple OR conditions?
350. How do you detect queries causing high CPU utilization?
351. How do you optimize queries using IN lists efficiently?
352. How do you implement covering indexes for aggregation queries?
353. How do you monitor query cache hit/miss ratios?
354. How do you optimize queries using temporary tables?
355. How do you rewrite queries to avoid correlated subqueries?
356. How do you detect and optimize queries with LIKE '%...%' patterns?
357. How do you optimize queries on TEXT/BLOB columns?
358. How do you monitor long-running queries in production?
359. How do you optimize queries for partitioned tables?
360. How do you detect queries causing deadlocks?
361. How do you optimize JOIN queries to minimize disk I/O?
362. How do you tune queries accessing multiple large tables simultaneously?
363. How do you implement query logging to detect performance regressions?
364. How do you optimize aggregation queries with multiple stages?
365. How do you detect queries using unoptimized functions in WHERE clauses?
366. How do you tune queries using derived tables or sub-selects?
367. How do you optimize queries using EXISTS vs IN?
368. How do you detect queries causing table scans on sharded data?
369. How do you optimize queries using proper JOIN order?
370. How do you implement query hints to enforce index usage?
371. How do you optimize queries for read-heavy reporting workloads?
372. How do you detect queries causing lock contention?
373. How do you optimize queries with multiple GROUP BY clauses?
374. How do you tune queries using LIMIT and OFFSET efficiently?
375. How do you detect queries causing replication lag?
376. How do you optimize queries using temporary result sets?

377. How do you monitor query execution plans over time?
378. How do you optimize queries for mixed OLTP/OLAP workloads?
379. How do you implement query profiling for production workloads?
380. How do you detect performance regressions after schema changes?

High Availability, Galera Clusters & Multi-Master (381–420)

381. How do you set up a MariaDB Galera cluster?
382. How do you monitor Galera cluster health?
383. How do you detect quorum loss in Galera?
384. How do you handle node desynchronization in Galera clusters?
385. How do you perform rolling upgrades in Galera clusters?
386. How do you optimize Galera for write-intensive workloads?
387. How do you detect and resolve deadlocks in Galera clusters?
388. How do you monitor replication conflicts in Galera?
389. How do you add a new node to a Galera cluster safely?
390. How do you remove a node without affecting cluster availability?
391. How do you detect and resolve SST (State Snapshot Transfer) issues?
392. How do you monitor Galera transaction certification performance?
393. How do you configure Galera cluster for multi-datacenter deployments?
394. How do you detect network latency affecting cluster performance?
395. How do you handle network partitions in Galera clusters?
396. How do you monitor cluster size and membership changes?
397. How do you configure Galera flow control to prevent overload?
398. How do you optimize Galera for high-concurrency reads?
399. How do you detect Galera nodes falling behind during replication?
400. How do you implement automated failover in multi-master setups?
401. How do you monitor Galera cluster for split-brain scenarios?
402. How do you tune wsrep_slave_threads for optimal performance?
403. How do you implement backup strategies in Galera clusters?
404. How do you detect write-set conflicts during heavy writes?
405. How do you optimize node join time during SST/IST?
406. How do you monitor Galera transaction latency?
407. How do you detect memory and disk pressure on cluster nodes?
408. How do you optimize cluster flow control to reduce stalling?
409. How do you implement monitoring dashboards for Galera?
410. How do you integrate Galera monitoring with Prometheus/Grafana?
411. How do you handle schema changes safely in Galera clusters?
412. How do you detect stale nodes affecting cluster consistency?
413. How do you optimize write load distribution in multi-master clusters?
414. How do you monitor Galera node uptime and performance metrics?
415. How do you implement automated node provisioning for Galera?
416. How do you tune innodb_flush_log_at_trx_commit in Galera?
417. How do you monitor cluster-wide performance trends?

418. How do you detect high replication latency across nodes?
419. How do you optimize Galera cluster for read-heavy workloads?
420. How do you implement failover testing for Galera clusters?

Monitoring, Security & Automation (421–460)

421. How do you monitor MariaDB query performance in real-time?
422. How do you detect and alert on slow queries automatically?
423. How do you implement centralized logging for multiple MariaDB servers?
424. How do you monitor disk and memory usage across servers?
425. How do you implement alerting for replication lag?
426. How do you monitor deadlocks in production environments?
427. How do you implement automated security audits?
428. How do you enforce SSL/TLS connections for all users?
429. How do you audit user privileges and activity?
430. How do you detect brute-force login attempts?
431. How do you implement automated backup and restore verification?
432. How do you monitor Galera SST/IST events?
433. How do you detect excessive connection usage?
434. How do you implement automated failover testing?
435. How do you integrate monitoring with Prometheus/Grafana?
436. How do you implement alerting for disk space thresholds?
437. How do you detect anomalies in query execution patterns?
438. How do you automate schema change verification?
439. How do you implement automated index optimization recommendations?
440. How do you monitor slow transactions and lock contention?
441. How do you detect failed replication events automatically?
442. How do you implement security alerts for privilege escalation?
443. How do you monitor query cache performance?
444. How do you automate node provisioning in production?
445. How do you monitor connection pooling and wait times?
446. How do you implement automated patch management?
447. How do you monitor SSL/TLS certificate expiration?
448. How do you detect and alert on high CPU/memory usage?
449. How do you implement monitoring dashboards for multiple clusters?
450. How do you detect stale or inactive nodes in Galera?
451. How do you automate backups for multi-datacenter environments?
452. How do you monitor performance regression after upgrades?
453. How do you detect query spikes in real-time?
454. How do you implement role-based monitoring access?
455. How do you detect failed login attempts and lock accounts automatically?
456. How do you monitor temporary table usage for performance alerts?
457. How do you automate security compliance reporting?
458. How do you monitor storage engine statistics?

459. How do you detect table fragmentation automatically?
460. How do you implement alerting for high disk I/O events?

Advanced Troubleshooting & Real-Time Scenarios (461–500)

461. How do you detect and resolve replication loops?
462. How do you troubleshoot Galera cluster desynchronization?
463. How do you resolve SST failures in Galera?
464. How do you troubleshoot slow query performance after schema changes?
465. How do you detect disk bottlenecks impacting InnoDB performance?
466. How do you resolve deadlocks in high-concurrency environments?
467. How do you troubleshoot high CPU usage by MariaDB processes?
468. How do you detect and fix query performance regressions after upgrades?
469. How do you troubleshoot replication conflicts in multi-master setups?
470. How do you detect lock wait time issues automatically?
471. How do you troubleshoot temporary table disk spills?
472. How do you detect network latency affecting replication?
473. How do you troubleshoot slow aggregation queries?
474. How do you detect inefficient queries using EXPLAIN and performance schema?
475. How do you optimize queries causing high I/O load during backups?
476. How do you troubleshoot connection pool bottlenecks?
477. How do you detect memory leaks in MariaDB processes?
478. How do you resolve index-related performance issues?
479. How do you troubleshoot query performance on partitioned tables?
480. How do you detect and resolve stale statistics causing slow queries?
481. How do you troubleshoot deadlocks in Galera clusters?
482. How do you detect and fix replication lag caused by heavy writes?
483. How do you troubleshoot InnoDB buffer pool contention?
484. How do you detect and fix queries performing full table scans unnecessarily?
485. How do you troubleshoot replication errors due to binary log corruption?
486. How do you detect and fix lock contention in high-concurrency OLTP workloads?
487. How do you troubleshoot SST/IST transfer failures in Galera?
488. How do you detect and optimize queries with heavy temporary table usage?
489. How do you troubleshoot slow queries in production without downtime?
490. How do you detect and fix high latency in multi-master replication?
491. How do you troubleshoot performance degradation after configuration changes?
492. How do you detect missing or unused indexes causing slow queries?
493. How do you troubleshoot Galera cluster flow control issues?
494. How do you detect replication inconsistency after failover?
495. How do you troubleshoot queries causing high memory usage?
496. How do you detect inefficient joins and rewrite them for performance?
497. How do you troubleshoot slow writes in replication setups?
498. How do you detect and fix performance bottlenecks caused by temporary tables?
499. How do you troubleshoot sudden disk I/O spikes affecting MariaDB?

Praveen Madupu
Mb: +91 98661 30093
Sr. Database Administrator

500. How do you plan and implement a MariaDB environment to scale for 10x growth while maintaining HA, performance, and security?

Above questions cover:

- **Basic Administration**
- **Query Performance & Optimization**
- **Indexing**
- **Replication & Galera Clusters**
- **Backup & Recovery**
- **Memory, Storage & Performance Tuning**
- **Monitoring, Security & Automation**
- **Advanced Troubleshooting**